

NUCLEUS

Centre of Excellence

JEE ADVANCED PATTERN

TEST-11 (29-09-2021)

Time: 3 Hours

OPEN TEST SERIES

Maximum Marks: 198

PAPER-2 WITH SOLUTION

READ THE INSTRUCTIONS CAREFULLY

GENERAL

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. The paper CODE is printed on the right hand top corner of this sheet and the right hand top corner of the back cover of this booklet.
3. Use the Optical Response Sheet (ORS) provided separately for answering the questions.
4. The paper CODE is printed on the left part as well as the right part of the ORS. Ensure that both these codes are identical and same as that on the question paper booklet. If not, contact the invigilator for change of ORS.
5. Blank spaces are provided within this booklet for rough work.
6. Write your name, roll number and sign in the space provided on the back cover of this booklet.
7. After breaking the seal of the booklet verify that the booklet contains **27 pages** and that all the **54 questions** along with the options are legible. If not, contact the invigilator for replacement of the booklet.
8. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET

9. Darken the appropriate bubbles on the ORS by applying sufficient pressure. This will leave an impression at the corresponding place on the Candidate's Sheet.
10. The ORS will be collected by the invigilator at the end of the examination.
11. You will be allowed to take away the Candidate's Sheet at the end of the examination.
12. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
13. Write your name, roll number and code of the examination center, and sign with pen in the space provided for this purpose on the ORS. Do not write any of these details anywhere else on the ORS. Darken the appropriate bubble under each digit of your roll number.

DARKENING THE BUBBLES ON THE ORS

14. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
15. Darken the bubble  **COMPLETELY**.
16. The correct way of darkening a bubble is as: 
17. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
18. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.

Please see the last page of this booklet for rest of the instructions.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR.

Name of the Candidate

I have read all the instructions and shall abide by them.

Signature of the Candidate

Form Number

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I have verified all the information filled in by the Candidate.

Signature of the invigilator

CHOOSE WISE TO RISE

Some Useful Data

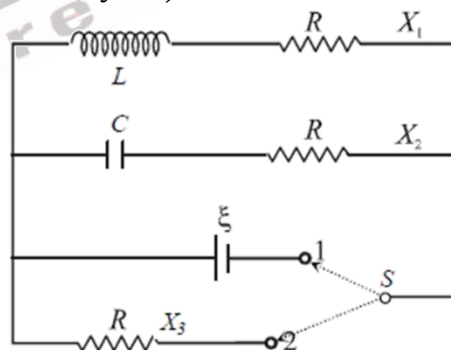
Quantity	Values
Constant of gravitation	$6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
Speed of light in vacuum	$2.99792458 \times 10^8 \text{ m s}^{-1}$
Avogadro constant	$6.0221367 \times 10^{23} \text{ mol}^{-1}$
Gas constant	$8.314510 \text{ JK}^{-1}\text{-mol}^{-1}$
Boltzmann constant	$1.380658 \times 10^{-23} \text{ JK}^{-1}$
	$8.617385 \times 10^{-5} \text{ eV K}^{-1}$
Stefan-Boltzmann constant	$5.67051 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$
Wien's displacement law constant	$2.897756 \times 10^{-3} \text{ m K}$
Charge of proton	$1.60217733 \times 10^{-19} \text{ C}$
Mass of electron	$9.1093897 \times 10^{-31} \text{ kg}$
	$5.48579903 \times 10^{-4} \text{ u}$
Mass of proton	$1.6726231 \times 10^{-27} \text{ kg}$
	1.007276470 u
Permeability of vacuum	$4\pi \times 10^{-7} \text{ NA}^{-2}$
Permittivity of vacuum	$8.854187817 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$
Faraday constant	$96485.3029 \text{ C mol}^{-1}$
Planck constant	$6.6260755 \times 10^{-34} \text{ J-s}$
	$4.1356692 \times 10^{-15} \text{ eV-s}$

PART-I : PHYSICS

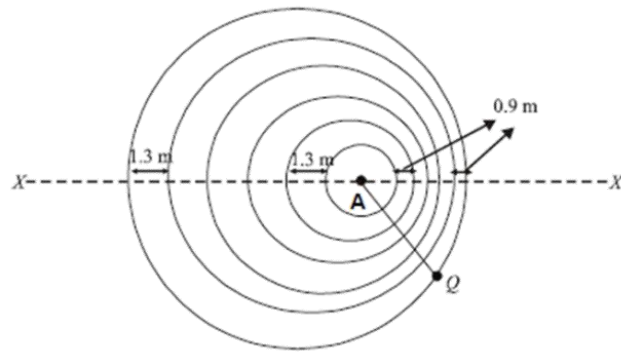
SECTION-I : Integer Value Correct Type

This section contains **6 questions**. The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, **BOTH INCLUSIVE**. using the mouse and the on screen virtual numeric keypad in the place designated to enter the answer. You will be awarded **3 marks** if correct numerical value is entered as answer. **(-1)** marks will be awarded for incorrect answers in this section.

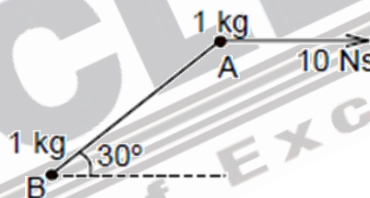
1. A perfectly absorbing surface intercepts a parallel beam of monochromatic light of power 100 W ($\lambda = 500$ nm) incident on it normally. If the force exerted by light beam on the surface is $\frac{10^{-6}}{n}$ N, then $n =$
2. In a moving coil galvanometer, torque on the coil can be expressed as $\tau \propto i$, where i is current through the wire. The rectangular coil of the galvanometer having number of turns N , area A and moment of inertia I is placed in magnetic field B . The torsional constant of spring in the galvanometer is C and charge Q is passed through the coil almost instantaneously. If the maximum angle through coil deflected is $\frac{KNABQ}{\sqrt{4CI}}$ (Ignore the damping in mechanical oscillation).
3. Figure shows a circuit containing three resistor X_1 , X_2 & X_3 having resistance R each, an inductor, capacitor and an emf source having inductance ' L ', capacitor C & emf ξ respectively (Given $R = \sqrt{\frac{L}{C}}$). The switch is first connected to position-1. when charge on capacitor becomes half of it maximum possible value then switch S is connected to position-2. Current in resistance X_3 just after shifting the switch from position-1 to position-2, is (given $L = 5\text{mH}$, $C = 2\mu\text{F}$, $R = 10\Omega$, $\xi = 5\text{V}$)



4. Figure shows wave fronts coming from a source at equal interval of $\frac{10}{3}$ milli second Frequency of wave emitted by source is 300 Hz. XX' is line along common diameter of wave fronts passing through 'A'. Distance between two consecutive wave fronts along line XX' is 0.9 m and 1.3 m respectively. At some instant source is at point A. Detector is placed at point 'Q' AQ makes an angle 60° with line XX' If frequency of wave received by detector is f Hz, then $\frac{f}{110}$ (in Hz) is equal to



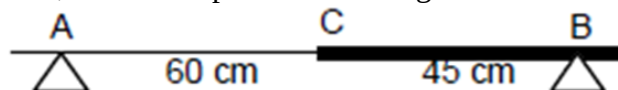
5. A rocket of mass 40kg had 360 kg of fuel. The exhaust velocity of the fuel is 2.0 km/sec. Calculate minimum rate of consumption of fuel in kg/s so that the rocket may rise from the ground ($g=10 \text{ m/sec}^2$)
6. Two balls A and B of masses 1 kg each are connected by an inextensible massless string. The system is resting on a smooth horizontal surface. An impulse of 10 Ns is applied to the ball A at an angle 30° with the line joining two balls in horizontal direction as shown in the figure. Assuming that the string remains taut after the impulse, if the magnitude of kinetic energy of ball A is $\frac{175}{k}$. Find the value of K.



SECTION-II : One or More Options Correct Type

This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE THAN ONE** may be correct. You will be awarded **4 marks** if all the correct option(s) is/are chosen and zero mark if none of the option is chosen. For each correct answer **+1** mark will be awarded if no incorrect option is chosen. In all other cases minus two (**-2**) mark will be awarded.

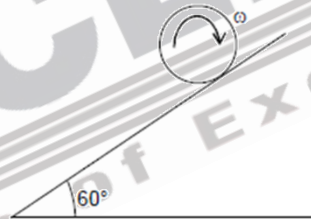
7. A steel wire of length 60 cm and area of cross section 10^{-6} m^2 is joined with an aluminium wire of length 45 cm and area of cross section $3 \times 10^{-6} \text{ m}^2$. The composite string is stretched by a tension of 80 N. Density of steel is 7800 kg m^{-3} and that of aluminium is 2600 kg m^{-3} . The minimum frequency of turning fork, which can produce standing wave in it with node at joint is



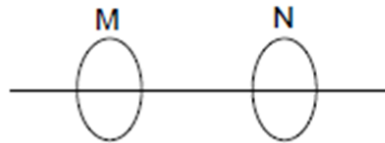
- (A) 357.3 Hz
(C) 337.5 Hz

- (B) 375.3 Hz
(D) 325.3 Hz

8. A point mass of 1 kg collides elastically with a stationary point mass of 5 kg. After their collision, the 1 kg mass reverses its direction and moves with a speed of 2 ms⁻¹. Which of the following statement(s) is (are) correct for the system of these two masses?
 (A) Total momentum of the system is 3 kg ms⁻¹
 (B) Momentum of 5 kg mass after collision is 4 kg ms⁻¹
 (C) Kinetic energy of the centre of mass is 0.75 J
 (D) Total kinetic energy of the system is 4 J
9. A strip of wood of mass M and length ℓ is placed on a smooth horizontal surface. An insect of mass m starts at one end on the strip and walks to the other end in time t, moving with a constant speed.
 (A) the speed of the insect as seen from the ground is $< \frac{\ell}{t}$
 (B) the speed of the strip as seen from the ground is $\frac{\ell}{t} \left(\frac{M}{M+m} \right)$
 (C) the speed of the strip as seen from the ground is $\frac{\ell}{t} \left(\frac{m}{M+m} \right)$
 (D) the total kinetic energy of the system is $\frac{1}{2} (m + M) \left(\frac{\ell}{t} \right)^2$
10. A cylinder is rotated clockwise and lowered slowly on a rough inclined plane with ($\mu = 0.8$). Then

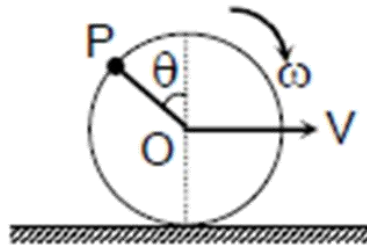


- (A) cylinder will start going upwards
 (B) cylinder will start going downwards.
 (C) frictional force will act upwards
 (D) frictional force will act downwards.
11. Two identical circular coils M and N are arranged coaxially as shown in the figure. Separation between the coils is large as compared to their radii. The arrangement is viewed from left along the common axis. The sign convention adopted is that currents are taken to be positive when they appear to flow in clockwise direction. Then



- (A) If M carries a constant positive current and is moved towards N, a positive current is induced in N
 (B) If M carries a constant positive current and N is moved towards M, a negative current is induced in N
 (C) If a positive current in M is switched off, a positive current is momentarily induced in N
 (D) If both coils carry positive currents, they will attract each other

12. A disc of radius R rolls on a horizontal surface with linear velocity V and angular velocity ω . There is a point P on circumference of disc at angle θ with upward vertical diameter measured anticlockwise see figure, which has a vertical velocity. Here θ is equal to

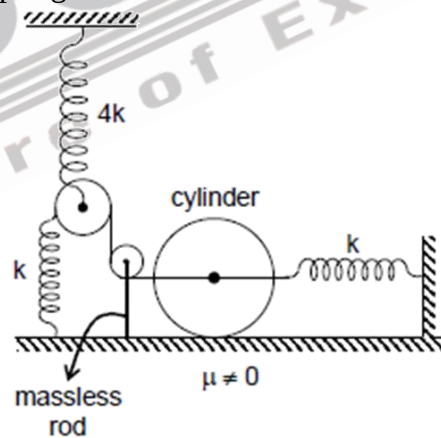


- (A) $\pi + \sin^{-1} \frac{V}{R\omega}$ (B) $\frac{\pi}{2} - \sin^{-1} \frac{V}{R\omega}$ (C) $\pi - \cos^{-1} \frac{V}{R\omega}$ (D) $\pi + \cos^{-1} \frac{V}{R\omega}$

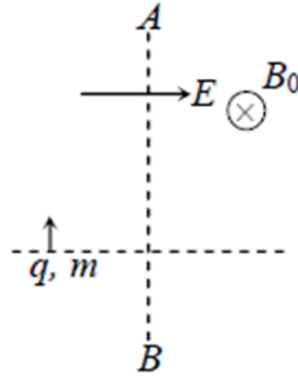
SECTION-III : Integer Value Correct Type

This section contains **6 questions**. For each question, enter the correct numerical value (in decimal notation, truncated / rounded off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30) using the mouse and the on screen virtual numeric keypad in the place designated to enter the answer. You will be awarded **4 marks** if correct numerical value is entered as answer. **No negative marks** will be awarded for incorrect answers in this section.

13. A cylinder of mass $m = 3$ kg and radius $R = 2$ m is attached to a massless spring-pulley system as shown in the figure and all the springs are in their natural length at $t = 0$. The friction is sufficient to cause pure rolling of the cylinder. What is the time period (in sec.) of cylinder for small horizontal displacement? (spring constant $k = 2$ N/m and $\pi = 3.14$)



14. A uniform electric field E is present horizontally along the paper throughout region but uniform magnetic field B_0 is present horizontally (perpendicular to plane of paper in inward direction) right to the line AB as shown. A charge particle having charge q and mass m is projected vertically upward and crosses the line AB after time t_0 . The speed of projection if particle moves after t_0 with constant velocity, (given $qE = mg$) is nt_0 . Find the value of n . ($g = 9.8$ m/s²)



15. A drop of water of mass $m = 0.4 \text{ g}$ is placed between two clean glass plates, the distance between which is 0.01 cm . If the force needed to hold one of the plates is $3.2 \times \text{N}$, find x' . (Surface tension of water = 0.08 N/m).
16. A bus is moving towards a huge wall with a velocity of 5 m/s . the driver sounds a horn of frequency 200 Hz . What is the frequency of beats heard by a passenger of the bus, if the speed of sound in air is 330 m/s ?
17. Two blocks of masses 6 kg and 4 kg connected by a rope of mass 4 kg are resting on frictionless floor as shown. If a constant force of 21 Newton is applied to 6 kg block, tension (In N) in the rope at point B is ($\text{CB} : \text{BA} = 1 : 1$)



18. Potential energy of a particle moving along x -axis is given by $U = \frac{x^3}{3} - \frac{9x^2}{2} + 20x$. Find out position of stable equilibrium state.

PART-II : MATHSMATICS

SECTION-I : Integer Value Correct Type

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19. If $n \in \mathbb{N}$, $n > 4$ and $(1-x^3)^n = \sum_{r=0}^n a_r x^r (1-x)^{3n-2r}$, then $\sum_{r=0}^n a_r = \lambda^n$, find λ .
20. 8 players $P_1, P_2, \dots, P_8$ of equal strength play in a knockout tournament. Assuming that players in each round are paired randomly. Then the probability that the player P_1 losses to the eventual winner is $\frac{m}{n}$, where m, n are relatively prime natural numbers, then $n-m$ is equal to
21. A normal to the hyperbola $\frac{x^2}{4} - \frac{y^2}{1} = 1$ has equal intercepts on positive x and y -axes. If this normal touches the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ then find $\frac{9}{25} (a^2 + b^2)$.
22. Let $\lim_{n \rightarrow \infty} n^{\frac{-1}{2} \left(1 + \frac{1}{n}\right)} \cdot (1^1 \cdot 2^2 \cdot 3^3 \cdot \dots \cdot n^n)^{\frac{1}{n^2}} = e^{\frac{-p}{q}}$ where p and q are relative prime positive integers. Find the value of $|p + q|$.
23. A conical vessel is to be prepared out of a circular sheet of metal of unit radius. In order that the vessel has maximum volume, the sectorial area that must be removed from the sheet is A_1 and the area of the given sheet is A_2 . If $\frac{A_2}{A_1} = m + \sqrt{n}$, where $m, n \in \mathbb{N}$, then $m + n$ is equal to
24. If $f(x) = \begin{cases} \cos x^3 & ; x < 0 \\ \sin x^3 - |x^3 - 1| & ; x \geq 0 \end{cases}$ then find the number of points where $g(x) = f(|x|)$ is non-differentiable.

SECTION-II : One or More Options Correct Type

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25. Let A is a non singular matrix such that $3ABA^{-1} + A = 2A^{-1}BA$, then
 (A) $A + B$ is singular matrix (B) $A + B$ is non singular matrix
 (C) $ABA^{-1} - A^{-1}BA$ is non singular matrix (D) $ABA^{-1} - A^{-1}BA$ is singular matrix

26. The equation $z^2 + \alpha z + \beta = 0$, where α, β are complex numbers has a
 (A) real root, if $(\bar{\alpha} - \alpha)(\alpha\bar{\beta} - \bar{\alpha}\beta) = (\beta - \bar{\beta})^2$
 (B) real root if $(\bar{\alpha} - \alpha)(\alpha\bar{\beta} + \bar{\alpha}\beta) = (\beta - \bar{\beta})^2$
 (C) a purely imaginary root if $(\alpha + \bar{\alpha})(\alpha\bar{\beta} + \bar{\alpha}\beta) + (\beta - \bar{\beta})^2 = 0$
 (D) a purely imaginary root if $(\alpha + \bar{\alpha})(\alpha\bar{\beta} - \bar{\alpha}\beta) + (\beta - \bar{\beta})^2 = 0$
27. Let $A = \{1, 2, 3, 4\}$, $B = \{1, 2, 3, 4, 5, 6, 7, 8\}$ then which of the following is true.
 (A) The number of functions $f : B \rightarrow A$, such that if $x_1 > x_2$, then $f(x_1) \geq f(x_2) \forall x_1 \in B$ is N. Then the sum of digits of N is 12.
 (B) The number of onto functions $f : B \rightarrow A$ such that if $x_1 > x_2$, then $f(x_1) \leq f(x_2) \forall x_1 \in B$ is N. Then the sum of digits of N is 8.
 (C) The numbers of one-one function $f : A \rightarrow B$ such that $f(x) \neq x \forall x \in A$ is N. Then the sum of digits of N is 8.
 (D) The number of one-one function $f : A \rightarrow B$ such that there are exactly two, $x, x \in A$ for which $f(x) = x$ is N. Then the sum of digits of N is 9.
28. The equation $x^4 - 9x^3 + 2(10-a)x^2 + 9ax + a^2 = 0$ for x where 'a' is real parameter has
 (A) 4 distinct real roots if $a > -4, a \neq 0$.
 (B) 2 distinct real roots if $-\frac{25}{4} < a < -4$
 (C) No real roots if $a < -\frac{25}{4}$.
 (D) 3 distinct real roots for 2 values of a.
29. It is given that the sequence $\{a_n\}$ satisfies $a_1 = 0, a_{n+1} = a_n + 1 + 2\sqrt{1+a_n}$ for $n \in \mathbb{N}$. Then
 (A) $a_{100} = 9999$ (B) $a_{2001} = 4004000$
 (C) $a_{2001} = 4002000$ (D) $a_{19} = 360$
30. Let $A(4, -2)$ and $B(0, 1)$ are two points on a parabola. Normals at A & B to the parabola intersect at $P(2, 2)$. Tangent parallel to AB to the parabola touches the parabola at Q and tangents drawn at A & B to the parabola meet at R.
 (A) Equation of directrix of the parabola is $2x + y + 1 = 0$
 (B) Equation of directrix of the parabola is $y = -3$
 (C) Length of latus rectum of parabola is $\frac{8}{5}$
 (D) Area of triangle AQB is equal to $\frac{5}{2}$

SECTION-III : Integer Value Correct Type

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31. The equation of plane which passes through the line of intersection of the planes $x + y + z - 1 = 0$ and $x - y + z - 3 = 0$ and which is at maximum distance from the point $(3, 4, 5)$ is $ax + by + cz = d$ then the value of $\frac{a+b-c}{4d}$ is
32. The number of solutions of the system of equations:

$$2\sin^2 x + \sin^2 2x = 2$$

$$\sin 2x + \cos 2x = \tan x$$
in $[0, 4\pi]$ satisfying $2\cos^2 x + \sin x \leq 2$ is:
33. If $\hat{a}, \hat{b}$ and $\hat{c}$ are non-coplanar unit vectors such that $[\hat{a} \hat{b} \hat{c}] = [\hat{b} \times \hat{c} \hat{a} \times \hat{a} \times \hat{b}]$, then find the projection of $\hat{b} + \hat{c}$ on $\hat{a} \times \hat{b}$.
34. If circumradius of $\triangle ABC$ is 3 units and its area is 6 units and $\triangle DEF$ is formed by joining foot perpendiculars drawn from A, B, C on sides BC, CA, AB respectively. Find the perimeter of $\triangle DEF$.
35. If A is the matrix $\begin{bmatrix} 1 & -3 \\ -1 & 1 \end{bmatrix}$ then $A - \frac{1}{3}A^2 + \frac{1}{9}A^3 \dots + \left(-\frac{1}{3}\right)^n A^{n+1} + \dots \infty = \frac{3}{13} \begin{bmatrix} 1 & a \\ b & 1 \end{bmatrix}$ Find $\left| \frac{a}{b} \right|$.
36. The radical centre of the three circles is at the origin. The equations of the two of the circles are $x^2 + y^2 = 1$ and $x^2 + y^2 + 4x + 4y - 1 = 0$. If the third circle passes through the points $(1, 1)$ and $(-2, 1)$; and its radius can be expressed in the form of $\frac{p}{q}$, where p and q are relatively prime positive integers. Find the value of $(p+q)$.

PART-III : CHEMISTRY

SECTION-I : Integer Value Correct Type

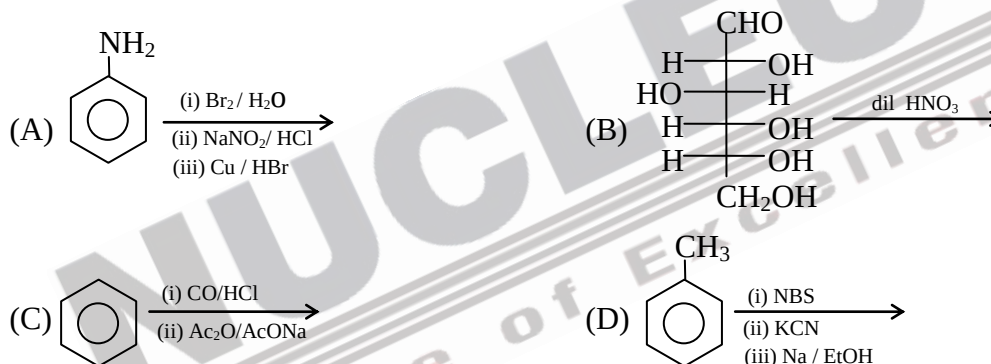
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37. Density of a Vander Waal's gas at 500 K and 1 atm is found to be 0.8 kg/m^3 . It is also found that the gas diffuses $\sqrt{\frac{3}{2}}$ times slower than pure O_2 gas under identical conditions. The value of $2 \times Z$ (where Z is compressibility factor of gas) for the real gas is ($R = 0.08 \text{ atm-L/mol-K}$) :

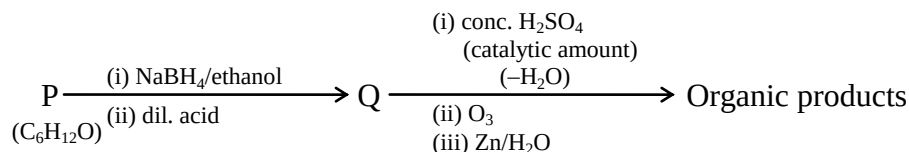
38. Find the number of planarmolecule.

(i) BF_3 (ii) BCl_3 (iii) CO_3^{2-} (iv) SO_3 (v) NH_3 (vi) NCl_3
(vii) PCl_3 (viii) PF_3

39. Major product of how many reactions are soluble in hot aqueous NaOH?



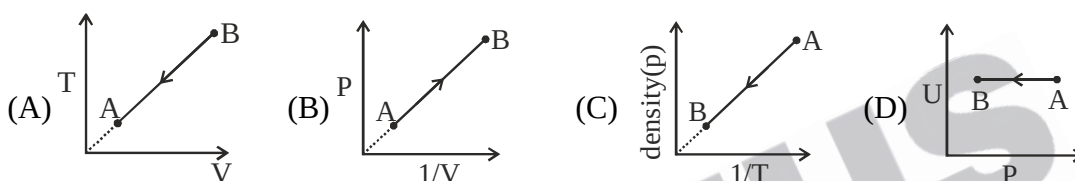
40. In He^+ ion sample electrons make transition from 5th excited state to 2nd state only, all different types of photons are observed. Then change in angular momentum of electrons producing 3rd highest energy photon is how much times of angular momentum of electron in first orbit.
41. Find the number of hypervalent compounds.
 BF_3 , SF_4 , NCl_3 , PCl_5 , SiF_4 , BrF_5 , XeF_2 , BeCl_2
42. P gave acetone as the one organic product through the following sequence of reactions, in which Q is an intermediate organic compound. Then how many P are possible :



SECTION-II : One or More Options Correct Type

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43. Which of the following is/are inner orbital complex(es) as well as diamagnetic in nature.
(A) $[\text{Ir}(\text{H}_2\text{O})_6]^{3+}$ (B) $[\text{Ni}(\text{NH}_3)_6]^{2+}$ (C) $[\text{Cr}(\text{NH}_3)_6]^{3+}$ (D) $[\text{Co}(\text{NH}_3)_6]^{3+}$
44. A sample of ideal gas is undergoing following process. For which of the following process $\Delta G \leq 0$?



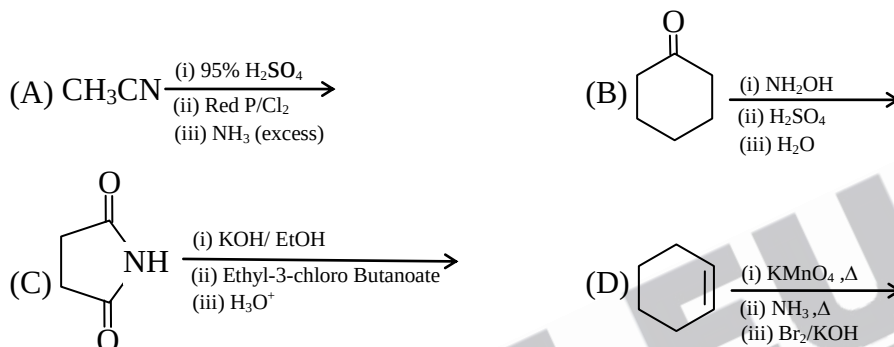
Select correct statement(s) :

- (A) $\Delta S_{\text{surrounding}}$ will be same for isothermal reversible and irreversible process for a system from same initial state to same final volume
(B) $\Delta_{\text{vapourisation}} G$ for water is zero at 1bar, 373K but $\Delta_{\text{vapourisation}} H$ is not.
(C) In an irreversible adiabatic process entropy of ideal gas always increases either its expansion or compression of gas.
(D) Activation energy of a chemical reaction can be determined by evaluating rate constant at two different temperatures
45. A hydro carbon X of molecular wt.54 reacts excess of Br_2 in CCl_4 to give a compound 'Y' whose molecular wt. is 374. However On catalytic hydrogenation with excess of hydrogen (X) forms (Z) whose molecular wt is only 7.4% more than that of (X), X reacts with $\text{CH}_3\text{CH}_2\text{Br}$ in presence of NaNH_2 to give another hydrocarbon 'P' which on ozonolysis yields diketone (Q). Then select the correct statements :
- (A) X on reaction with $\text{Ag}_2\text{O}/\text{OH}^-$ produces white precipitate.
(B) Z can be obtained in good yield by wurtz reaction.
(C) Q on reaction with m-CPBA following by hydrolysis produces propanoic acid.
(D) P on reaction with $\text{Na}/\text{liq. NH}_3$ followed by Br_2/CCl_4 produces meso compound.
46. Which metal impart(s) red colour to flame?
(A) Li (B) K (C) Ca (D) Sr
47. 12 ml of a gaseous mixture of an alkane & an alkene (containing same number of carbon atoms) require exactly 285 ml of air (containing 20% v/v O_2 and rest N_2) for complete combustion at 200

K. After combustion when gaseous mixture is passed through KOH solution it shows volume contraction of 36 ml. Which of the following is/are correct is :

- (A) Mole fraction of CO_2 in final gaseous sample is $\frac{6}{44}$
 (B) Formula of alkane is C_3H_8
 (C) Mole fraction of CO_2 in final gaseous sample is $\frac{6}{38}$
 (D) Total volume of CO_2 produced is 36 mL

48. Which of the following reaction will produce monomer of biodegradable polymers?



SECTION-III : Integer Value Correct Type

This section contains **6 questions**. For each question, enter the correct numerical value (in decimal notation, truncated / rounded off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30) using the mouse and the on screen virtual numeric keypad in the place designated to enter the answer. You will be awarded **4 marks** if correct numerical value is entered as answer. **No negative marks** will be awarded for incorrect answers in this section.

49. In solid ZnS , anions form fcc and cations are present in alternate tetrahedral void. If half of Zn^{2+} are replaced by Li^+ ions and some of the S^{2-} are replaced by Cl^- ions then find the empirical formula of the solid. If your answer is $\text{Li}_x\text{Zn}_y\text{S}_z\text{Cl}_m$ then fill your answer as $(x + y + z + m)$

50. In order to concentrate galena (which contains ZnS as impurity) by froth-floatation process, sodium cyanide is used as depressant. NaCN dissolves ZnS due to formation of water soluble complex(A).
Find the value of $w + x + y + z$.

where w = coordination number of central metal ion in complex ion of (A)

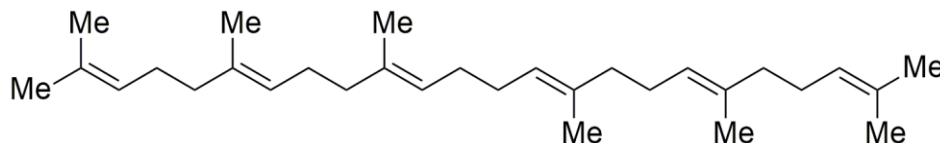
x = number of unpaired electrons in (A)

y = total number of possible linkage isomers of (A) including (A)

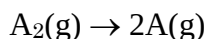
z = maximum number of atoms in a single plane in the complex ion of (A).

51. Find the value of $\left(\frac{Q \times R \times S}{P}\right)$

- (a) The dihedral (torsional) angle (in degrees) between the two methyl groups in the most stable conformation of n-butane is P (Round off to nearest integer)
 (b) Total number of constitutional isomers possible for trimethyl cyclohexane is Q
 (c) The degree of unsaturation (double bond equivalent) for a compound with molecular formula $C_{14}H_{12}O_2$ is R
 (d) The total number of head to tail isoprene linkages in the following molecule is S



52. A diatomic gas $A_2(g)$ decomposes into atomic $A(g)$ by first order kinetics as:



An empty container was filled with $A_2(g)$ and $N_2(g)$ at an initial pressure of 800 mm of Hg at 600 K and heated. After very long time, gases in the flask developed a pressure of 1400 mm of Hg. If half life for the decomposition process is 2 hr. What was the pressure (in mm Hg) in the flask after 4 hr from start. Assume N_2 to be an inert gas.

53. Consider the following equation



When compound (A) is allowed to react with P_2O_5 compound (C) is formed. How many $d_\pi-p_\pi$ bonds are present in compound (C).

Note: There are two separate parts in this question i.e. part (a) A part (b). Suppose answer for part (a) is P and for part (b) is Q, then final answer of this question will be the form of $P + Q$ in OMR Sheet.

54. Compound 'P' of molecular formula ' C_4H_{10} ' gives two monochlorinated structures on monochlorination. Treatment of both monochlorinated compounds with t-BuONa/t-BuOH gives only one alkene. Then sum of total stable conformer of compound 'P' and total α -H present in alkene is :

ANSWER KEY & SOLUTION

PART-I : PHYSICS

1.	3	2.	2	3.	0	4.	3	5.	2	6.	8	7.	C
8.	AC	9.	AC	10.	BC	11.	BCD	12.	CD	13.	8.56	14.	19.60
15.	2.00	16.	6.00	17.	9.00	18.	5.00						

PART-II : MATHEMATICS

19.	4	20.	5	21.	3	22.	5	23.	9	24.	2	25.	AD
26.	AC	27.	ABD	28.	ABCD	29.	ABD	30.	BD	31.	1.25		
32.	8.00	33.	1.00	34.	4.00	35.	3.00	36.	5.00				

PART-III : CHEMISTRY

37.	3	38.	4	39.	3	40.	2	41.	4	42.	7	43.	AD
44.	BCD	45.	ABCD	46.	ACD	47.	ABD	48.	ABC	49.	4.00	50.	14.00
51.	1.20	52.	125.00	53.	3.33	54.	7.00						

SOLUTION | PART-I : PHYSICS | ADVANCED PAPER

1. (3)

Sol. $F = \frac{dP}{dt}$ = rate of change of momentum of the photons. Thus, in case of complete absorption, $F =$
Intensity of the wave / speed of light.

2. (2)

Sol. $\int \tau dt = NAB \int i dt$

$$I\omega_0 = NABQ \dots\dots\dots(i)$$

$$\frac{1}{2} I\omega_0^2 = \frac{1}{2} C\theta_{\max}^2 \dots\dots\dots(ii)$$

From equation (i) and (ii)

$$\theta_{\max} = \frac{NABQ}{\sqrt{CI}}$$

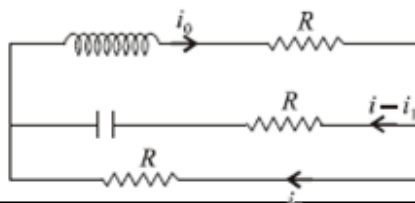
3. (0)

Sol. $R - \sqrt{\frac{L}{C}} \Rightarrow RC = L/R$ i.e. time constants of RC & LR circuit are same

After switching from 1 to 2

$$i_0 = \frac{\xi}{2R}$$

from loop rule,



$$-(i_0 - i_1)R + \varepsilon / 2 + i_1 R = 0$$

$$\Rightarrow i_1 = \frac{2i_0 R - \xi}{4R} = 0$$

4. (3)

Sol. $\lambda_0 - V_s t = x_1$

$$\lambda_0 - V_s t = x_2$$

$$V_s = \frac{x_2 - x_1}{2T} = 60 \text{ m/s} \text{ \& } \lambda_0 = \frac{x_1 + x_2}{2} = 1.1 \text{ m}$$

$$\text{Speed of sound } V = n \times \lambda_0 = 330 \text{ m/s}$$

Frequency detected at point Q

$$n = \frac{V}{V - V_s} \times 300 \text{ Hz} = 330 \text{ Hz}$$

5. (2)

Sol. We have relation $M \frac{dv}{dt} = F_{\text{ext}} + V_{\text{rel}} \frac{dM}{dt}$

In the question, we have the force by the gravity as the external force. As the rocket is to lift from the ground the minimum acceleration of the rocket to do so is, 0, beyond which, with a slight increase it will zoom into the sky.

Therefore, $\frac{dv}{dt} = 0$

Putting these into the equation, we obtain

$$0 = Mg + V_{\text{rel}} \left(-\frac{dM}{dt} \right)$$

$\left(\frac{dM}{dt} \right)$ is rate of decrease of mass hence negative. Mg is in the direction of the relative velocity of the fuel with respect to the rocket)

$$\text{Hence, } \frac{dM}{dt} = \frac{M}{V} g = \frac{400 \times 10}{2 \times 10^3} = 2 \text{ kg/sec}$$

6. (8)

Sol. Let us is the velocity of A and B along the string $10 \cos 30 = 2u$

$$\Rightarrow u = \frac{5\sqrt{3}}{2} \text{ m/s}$$

Let v is velocity of A $\perp$ to the string.

$$10 \sin 30 = v$$

$$\Rightarrow v = 5 \text{ m/s}$$

$$\therefore V_A = \sqrt{u^2 + v^2} = \frac{\sqrt{175}}{2}$$

$$KE_A = \frac{175}{8} \text{ J}$$

7. (C)

Sol. Mass per unit length, $\mu = \frac{m}{\ell} = \frac{SA\ell}{\ell} = \rho A$

$$\mu_s = \mu_{Al} = 78 \times 10^{-4} \text{ kg/m}$$

$\therefore$ Speed of wave is same in both wire

$$V = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{80 \times 10^4}{78}} = \frac{2 \times 10^2}{\sqrt{3.9}}$$

$$V_{\min} = \frac{V}{\lambda_{\max}} = \frac{200}{\sqrt{3.9} \times 0.3} \left[\frac{\lambda_{\max}}{2} = 15 \text{ cm for C as a node} \right]$$

$$337.4 \text{ Hz}$$

8. (A,C)

Sol. $e = \frac{v_2 - v_1}{u_1 - u_2}$

$$1 = \frac{v_2 - (-2)}{u_1}$$

$$u_1 = v_2 + 2$$

$$u_1 = 1(-2) + 5(u_1 - 2)$$

$$u_1 = -2 + 5u_1 - 10$$

$$u_1 = \frac{12}{4} = 3 \text{ m/s}$$

$$v_2 = 1 \text{ m/s}$$

$$\text{Kinetic energy of the centre of mass} = \frac{1}{2} \times (1+5) \times \left(\frac{3}{1+5} \right)^2 = \frac{3}{4} \text{ J}$$

9. (A,C)

Sol. Velocity of insect w.r.t. strip = $\frac{\ell}{t}$

Let strip moves with speed v Initial momentum was 0

$$\Rightarrow 0 = m \left(\frac{\ell}{t} + v \right) + Mv$$

$$v = -\frac{m\ell/t}{M+m} \text{ i.e. } \frac{m\ell/t}{M+m} \text{ toward left}$$

$$\therefore \text{Velocity of insect w.r.t. ground} = \frac{\ell}{t} - \frac{m\ell/t}{M+m} < \frac{\ell}{t}$$

10. (B,C)

Sol. Cylinder will move downwards as $mg \sin \theta > \mu \cos \theta$ although frictional force acts upwards.

11. (B,C,D)

Sol. Apply Faraday law for direction of induced current.

12. (C,D)

Sol. Particle P will have a velocity in vertical direction.

i.e. $R\omega \cos \alpha = V$

$$\alpha = \cos^{-1} \frac{V}{R\omega}$$

Required angle θ is $\theta = \pi - \alpha = \pi - \cos^{-1} \frac{V}{R\omega}$

And in second case $R\omega \cos \alpha = V$

$$\alpha = \cos^{-1} \frac{V}{R\omega}, \theta = \pi + \cos^{-1} \frac{V}{R\omega}$$

13. (8.56)

Sol. When the cylinder is displaced towards right, (θ is angle of rotation of diameter of the cylinder about the contact point) then

$$x_1 + 2x_2 = R\theta$$

$$x_1 + x_1 = R\theta$$

$$x_1 = \frac{R\theta}{2}$$

$$\tau = -[kx_1 R + kR\theta R]$$

$$\frac{3}{2}mR^2\alpha = -\left[\frac{KR^2\theta}{2} + KR^2\theta\right]$$

$$\alpha = -\frac{k}{m}\theta \quad ; \quad t_1 = \pi\sqrt{\frac{m}{k}}$$

When cylinder moves towards left, then

Torque about the contact point

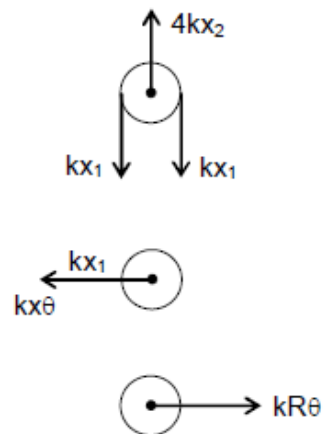
$$-kR^2\theta = \frac{3}{2}mR^2\alpha$$

$$\alpha = -\left(\frac{2}{3}\frac{k}{m}\right)\theta \quad ; \quad t_2 = \pi\sqrt{\frac{3m}{2k}}$$

So, time period of oscillation

$$T = t_1 + t_2$$

$$= \frac{\pi}{2}(3 + \sqrt{6}) = 5.56 \text{ sec.}$$



14. (19.60)

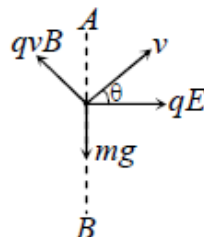
Sol. When crosses AB

$$qvB_0 \cos \theta = mg$$

$$qvB_0 \sin \theta = qE$$

$$\tan \theta = \frac{qE}{mg} = 1$$

$$\theta = \frac{\pi}{4}$$



along horizontal $v \cos \theta = \frac{qE}{m} t_0$

$u - gt_0 = v \sin \theta$

$u = \left(g + \frac{qE}{m} \right) t_0 = 2gt_0$

15. (2.00)

Sol. Let R be the radius of the circular layer of water.

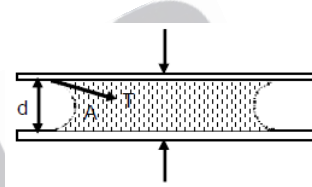
Then, $\pi R^2 d \times \rho = m$ (1)

Pressure at A = $p_0 - \frac{2T}{d}$ (2)

Thus pressure between the plates is less than the atmospheric pressure and so the plates are pressed together as through attracted towards each other.

F , force of attraction = $\Delta p \times \text{area} \Rightarrow F = \frac{2T}{d} \times \pi R^2$

$\Rightarrow F = \frac{2T}{d} \times \frac{m}{d \cdot \rho} = \frac{2T}{d^2 \cdot \rho} \times \frac{2 \times 0.4 \times 10^{-3} \times 0.08}{0.01^2 \times 10^{-4} \times 1000} = 6.4 \text{ N}$



16. (6.00)

Sol. As source (horn of bus) is approaching stationary wall (say, listener), therefore, apparent frequency striking the wall is

$v' = \frac{v v}{v - v_s}$ (1)

Sound of this frequency will be reflected by the wall (now, source). The passenger is the listener moving towards source. Therefore, frequency heard by the listener

$v'' = \frac{v + v_L}{v} \times \frac{v v}{v - v_s} = \frac{(v + v_L) v}{v - v_s}$

$= \frac{(330 + 5) \times 200}{330 - 5} = \frac{335}{325} \times 200$

$v'' = 206 \text{ Hz}$

$\therefore \text{Beat frequency} = (v'' - v)$

$= 206 - 200 = 6 \text{ Hz}$

17. (9.00)

Sol. $a = \frac{21}{14}$, $T_B = 6a = \frac{6 \times 21}{14} = 9 \text{ N}$.

18. (5.00)

Sol. As stable equilibrium, U is minimum. Thus $\frac{d^2U}{dx^2} > 0$.

$$\text{And } \frac{dv}{dx} = 0.$$

$$= \frac{1}{dx} \left(\frac{x^3}{3} - \frac{ax^2}{2} + 20x \right) = 0.$$

$$\Rightarrow x^2 - 9x + 20 = 0. \Rightarrow (x - 5)(x - 4) = 0.$$

$x = 5$ and $x = 4$ are point of equilibrium

And U minimum when $\frac{d^2U}{dx^2} > 0$. i.e. at $x = 5$.

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PART-II : MATHSMATICS

19. (4)

Sol. $(1-x^3)^n = \sum_{r=0}^n a_r x^r (1-x)^{3n-r}$

$$\Rightarrow (1-x^3 + 3x^2 - 3x^2 + 3x)^n$$

$$\Rightarrow \left[(1-x^3) + 3x(1-x) \right]^n = {}^nC_r (1-x)^{3(n-r)} (3x(1-x))^r$$

$$= {}^nC_r (1-x)^{3n-2r} \cdot x^r \cdot 3^r$$

$$ar = {}^nC_r 3^r$$

$$\text{Now } \sum a_r = (1+3)^n = 4^n \Rightarrow r = 4$$

20. (5)

Sol. P₁ Loses to winner

Winner can be selected in 7C_1 ways

$$\text{Req. probability} = \frac{7}{8} \times \frac{{}^6C_2}{{}^7C_3}$$

$$= \frac{7}{8} \times \frac{3}{7} = \frac{3}{8} = \frac{m}{n}$$

$$n - m = 5$$

21. (3)

Sol. $\frac{x^2}{4} - \frac{y^2}{1} = 1$ slope of normal = -1

$$\frac{2x}{4} - 2y \frac{dy}{dx} = 0 \quad \text{at}(x_1, y_1)$$

$$\Rightarrow \frac{dy}{dx} = \frac{x_1}{4y_1}$$

So $\frac{-4y_1}{x_1} = -1 \Rightarrow x_1 = 4y_1$

$$\Rightarrow 4y_1^2 - y_1^2 = 1 \Rightarrow y_1 = \frac{\pm 1}{\sqrt{3}}, x_1 = \pm \frac{4}{\sqrt{3}}$$

Eqⁿ of Normal $y - \frac{1}{\sqrt{3}} = -1 \left(x - \frac{4}{\sqrt{3}} \right)$

$$\Rightarrow x + y = \frac{5}{\sqrt{3}}$$

Eqⁿ of tangent to Ellipse is $y = mx \pm \sqrt{a^2 m^2 + b^2}$

for $m = -1$, $\sqrt{a^2 + b^2} = \frac{5}{\sqrt{3}}$

$$\Rightarrow a^2 + b^2 = \frac{25}{3}$$

$$\text{So } \frac{9}{25}(a^2 + b^2) = 3$$

22. (5)

Sol. $l = \lim_{n \rightarrow \infty} n^{\left(\frac{n+1}{2n}\right)} (1^1 - 2^2 \dots n^n)^{\frac{1}{n^2}}$

$$l = \lim_{n \rightarrow \infty} \left[\frac{1^1 \cdot 2^2 \dots n^n}{n^{\frac{n(n+1)}{2}}} \right]^{\frac{1}{n^2}}$$

$$l = \lim_{n \rightarrow \infty} \prod_{r=1}^n \left(\frac{r}{n} \right)^{\frac{r}{n^2}}$$

$$\text{Log } l = \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{r}{n^2} \log \left(\frac{r}{n} \right), \quad \frac{r}{n} = x, \quad \frac{1}{n} = dx$$

$$\text{Log } l = \int_0^1 x \log x dx$$

$$= \log x \cdot \frac{x^2}{2} - \int_0^1 \frac{x}{2} dx$$

$$\text{Log } l = 0 - \left[\frac{x^2}{4} \right]_0^1 = -\frac{1}{4}$$

$$\Rightarrow l = e^{-\frac{1}{4}}, \quad \frac{p}{q} = \frac{1}{4} \Rightarrow |p+q| = 5$$

23. (9)

Sol. Surface area = $\pi r l = \pi - A_1 r = 1 - \frac{A_1}{\pi} h^2 + r^2 = 1$

$$\Rightarrow h = \sqrt{1-r^2}$$

$$V = \frac{1}{3} \pi (1-h^2) h$$

$$\frac{dv}{dh} = \frac{\pi}{3} (1-3h^2) = 0 \Rightarrow h = \frac{1}{\sqrt{3}}, r = \frac{\sqrt{2}}{\sqrt{3}}$$

$$\frac{A_2}{A_1} = \frac{\sqrt{3}}{\sqrt{3}-\sqrt{2}}$$

$$= \sqrt{3}(\sqrt{3}+\sqrt{2})$$

$$= 3 + \sqrt{6}$$

$$= 9$$

24. (2)

Sol. $f(x) = \begin{cases} \cos x^3, & x < 0 \\ \sin x^3 - |x^3 - 1|, & x \geq 0 \end{cases}$

$f(x) = \begin{cases} \cos x^3, & x < 0 \\ \sin x^3 + x^3 - 1, & 0 \leq x < 1 \\ \sin x^3 + 1 - x^0, & 1 < x < \infty \end{cases}$

f is Cont. at $x = 0$ & 1 , but not differentiable at $x = 1$

hence $g(x)$ is not differentiable at $|x| = 1$

$$\Rightarrow x = \pm 1$$

25. (A,D)

Sol. $3A^3B^{-1} + A = 2A^{-1}BA$
 $\Rightarrow 3AB + A^2 = 2A^{-1}BA^2$
 $\Rightarrow 3A^2B + A^3 = 2BA^2$
 $\Rightarrow 3A^2B + 3A^3 = 2BA^2 + 2A^3$
 $\Rightarrow 3A^2(B+A) = 2(B+A)A^2$
 So $3^n|A|^2|B+A| = 2^n|B+A||A|^2$
 So $|B+A| = 0$
 Also $3ABA^{-1} - 3A^{-1}BA = -A^{-1}BA - A$
 $\Rightarrow 3A(ABA^{-1} - A^{-1}BA) = -BA - A^2$
 $3A(ABA^{-1} - A^{-1}BA) = -(B+A)A$
 Since $|A+B| = 0$, so
 $|ABA^{-1} - A^{-1}BA| = 0$

26. (A,C)

Sol. $Z^2 + \alpha z + \beta = 0$
 If Eqⁿ has real roots then
 $z^2 + \alpha z + \beta = 0$
 $z^2 + \bar{\alpha}z + \bar{\beta} = 0$ have a common root
 So $(\bar{\alpha} - \alpha)(\alpha\bar{\beta} - \bar{\alpha}\beta) = (\bar{\beta} - \beta)^2$
 If Eqⁿ has purely imaginary root then
 $z^2 + \alpha z + \beta = 0$
 $z^2 - \bar{\alpha}z + \bar{\beta} = 0$ have common root
 $(-\bar{\alpha} - \alpha)(\alpha\bar{\beta} - \bar{\alpha}\beta) = (\bar{\beta} - \beta)^2$
 $\Rightarrow (\alpha + \bar{\alpha})(\alpha\bar{\beta} + \bar{\alpha}\beta) + (\beta - \bar{\beta})^2 = 0$

27. (A,B,D)

Sol. $B = \{1, 2, 3, 4, 5, 6, 7, 8\}$

$A = B = \{1, 2, 3, 4\}$

(A) No. of functions from B to A

$$\Rightarrow {}^{8+4-1}C_{4-1} = {}^{11}C_3 = 165 = N$$

Sum of digits of N = 12

(B) No. of onto functions

$$\Rightarrow {}^{8-1}C_{4-1} = {}^7C_3 = 35 = N$$

Sum of digits of N = 8

(D) No. of one-one functions from A to B for Exactly two $f(x) = x$ is ${}^4C_2 \times [4 \times 3 + {}^2C_1 \times 4 + 1]$

N = 126

Sum of digits = 9

28. (A,B,C,D)

Sol. $x^4 - 9x^3 + 2(10 - a)x^2 + 9ax + a^2 = 0$

$$\Rightarrow a^2 + a(9x - 2x^2) + x^4 - 9x^3 + 20x^2 = 0$$

$$\Rightarrow (a - (x^2 - 4x))(a - (x^2 - 5x)) = 0$$

$$a = x^2 - 4x \quad \text{or} \quad a = x^2 - 5x$$

$$\Rightarrow x^2 - 4x - a = 0 \quad \& \quad x^2 - 5x - a = 0$$

$$D_1 = 16 + 4a, \quad D_2 = 25 + 4a$$

For 4 distinct real roots, $a > -4$.

For 2 distinct real roots, $\Rightarrow a = -4, 0$

For No real root $a < \frac{-25}{4}$

29. (A,B,D)

Sol. $a_{n+1} = a_n + 1 + 2\sqrt{1 + a_n}$

$$\Rightarrow a_{n+1} + 1 = (\sqrt{a_{n+1}} + 1)^2$$

$$\Rightarrow \sqrt{a_{n+1} + 1} - \sqrt{a_n + 1} = 1$$

So $\sqrt{a_2 + 1} - \sqrt{a_1 + 1} = 1$

$$\sqrt{a_3 + 1} - \sqrt{a_2 + 1} = 1$$

.

.

.

$$\sqrt{a_n + 1} - \sqrt{a_{n-1} + 1} = 1$$

Adding all these we get

$$\Rightarrow \sqrt{a_n + 1} - \sqrt{a_1 + 1} = n - 1$$

$$\sqrt{a_n + 1} = n$$

So $a_n = n^2 - 1$

$$a_{100} = 9999, \quad a_{2001} = 400400$$

$$a_{19} = 360$$

30. (B,D)

Sol. $y = ax^2 + bx + c$

$$c = 1, \quad -2 = 16a + 4b + 1$$

$$\frac{dy}{dx} = 2ax + b, y - 1 = \frac{-1}{b}(x - 0)$$

$$b = -2$$

$$y = \frac{5}{16}x^2 - 2x + 1$$

31. (1.25)

Sol. Eqⁿ of plane through intersection of P₁ & P₂ & containing (3,4,5)

$$\Rightarrow x+y+z-1 + \lambda(x-y+z-3) = 0$$

$$\lambda = -1$$

$$\text{Eq}^n \text{ of plane } 5x - 6y + 5z - 16 = 0$$

Plane at maximum distance will be $\perp$ to this plane.

$$(x+y+z-1) + r(x-y+z-3) = 0$$

$$\Rightarrow 5(1+r) - 6(1-r) + 5(1-r) = 0$$

$$r = \frac{-1}{4}$$

$$\text{So } 3x + 5y + 3z = 1$$

$$\frac{a+b-c}{4d} = 1.25$$

32. (8.00)

Sol. $2\sin^2 x + \sin^2 2x = 2$

$$\Rightarrow \sin^2 2x = 2\cos^2 x$$

$$\Rightarrow \cos^2 x = 0, \sin^2 x = \frac{1}{2}$$

$$\text{So } x = (2n+1)\frac{\pi}{2} \text{ or } x = n\pi \pm \frac{\pi}{4}$$

$$\Rightarrow \sin 2x + \cos 2x = \tan x$$

$$\text{for } x = n\pi \pm \frac{\pi}{4}$$

$$\text{Also } 2\cos^2 x + \sin x = 1 \pm \frac{1}{\sqrt{2}} \text{ for}$$

$$x = n\pi \pm \frac{\pi}{4}, \text{ so total 8 Solutions.}$$

33. (1.00)

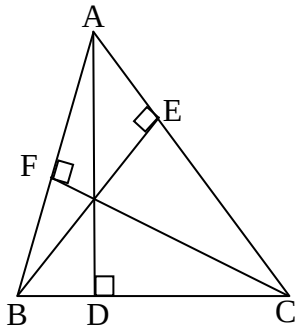
Sol. $[\vec{a} \vec{b} \vec{c}] = [\vec{a} \vec{b} \vec{c}]^2$

$$\text{So } [\vec{a} \vec{b} \vec{c}] = 1$$

$$\text{Now } \frac{(\hat{b} + \hat{c}) \cdot (\hat{a} \times \hat{b})}{|\hat{a} \times \hat{b}|} = [\hat{c} \hat{a} \hat{b}] = 1$$

34. (4.00)

Sol.



$$EF = R \sin 2A$$

$$FD = R \sin 2B$$

$$DE = R \sin 2C$$

$$\begin{aligned} \text{Perimeter} &= R. [\sin 2A + \sin 2B + \sin 2C] \\ &= 4R \sin A \sin B \sin C \\ &= \frac{2\Delta}{R} = \frac{2 \times 6}{3} = 4 \end{aligned}$$

35. (3.00)

Sol. $A = \begin{bmatrix} 1 & -3 \\ -1 & 1 \end{bmatrix}$

$$S = A - \frac{1}{3}A^2 + \frac{1}{9}A^3 \dots\dots$$

$$\frac{1}{3}AS = \frac{A^2}{3} - \frac{A^3}{9} \dots\dots$$

$$\Rightarrow \left(I + \frac{A}{3} \right) S = A$$

$$\Rightarrow \frac{S}{3} = (I + \frac{A}{3})^{-1} A$$

$$\Rightarrow \frac{S}{3} = \begin{bmatrix} 4 & -3 \\ -1 & 4 \end{bmatrix}^{-1} A$$

$$S = \frac{3}{13} \begin{bmatrix} 1 & -9 \\ -3 & 1 \end{bmatrix}$$

$$\left| \frac{a}{b} \right| = \frac{9}{3} = 3$$

36. (5.00)

Sol. $x^2 + y^2 = 1$
 $x^2 + y^2 + 4x + 4y - 1 = 0$

Eqn of third circle be

$$(x-1)(x+2) + (y-1)(y-1) + \lambda(y-1) = 0$$

distance from origin is (-1)

$$\Rightarrow -2 + 1 + \lambda(-1) = -1$$

$$\lambda = 0$$

Eqⁿ of circle is

$$x^2 + y^2 + x - 2y - 1 = 0$$

$$\text{radius} = \sqrt{\frac{1}{4} + 1 + 1} = \sqrt{\frac{9}{4}} = \frac{3}{2}$$

$$\frac{P}{q} = \frac{3}{2} \Rightarrow P + q = 5$$

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PART-III : CHEMISTRY

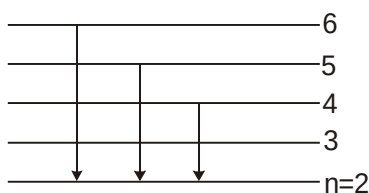
37. (3)

38. (4)

39. (3)

40. (2)

Sol.



3rd highest line is $4 \rightarrow 2$

Hence change in angular momentum which is 2 times of $\frac{h}{2\pi}$.

41. (4)

42. (7)

43. (A,D)

44. (B,C,D)

45. (A,B,C,D)

46. (A,C,D)

47. (A,B,D)

48. (A,B,C)

49. (4.00)

50. (14.00)

51. (1.20)

52. (125.00)

Sol. $t = 0$ 600
 $A_2 \rightarrow 2A$
 $T = 4 \text{ hr.}$ 150 900
Total pressure = $150 + 900 + 200$
 = 1250 mm Hg
 = 125 cm Hg

53. (3.33)

54. (7.00)

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QUESTION PAPER FORMAT AND MARKING SCHEME

19. The question paper has three parts: Physics, Mathematics and Chemistry.
20. Each part has three sections as detailed in the following table:

NAME OF THE CANDIDATE : _____

ROLL NO. : _____

I have read all the instructions
and shall abide by them

Signature of the Candidate

I have verified the identity, name and roll number
of the candidate.

Signature of the invigilator